

network adversary: reads, drops, replays, modifies, injects, stores for later

Eavesdropping
store now, decrypt later → **ML-KEM-1024** **X25519** **AES-256-GCM**

Impersonation
man in the middle → **ML-DSA-87** pinned keys, Finished MACs

Replay
duplicates, reordering → window of 2048

Tampering
modify, inject → **GCM tag** over header and payload

Downgrade
weaker algorithm suite → **ML-DSA-87** signs the suite identifier

Key compromise
past and future traffic → **epoch chain** **PQ ratchet** fresh secrets

Seized UAV
stored photos, video → **ML-KEM-1024** **X25519** wrap the file key

Forged file
planted recording → **ML-DSA-87** signs every stored file

Handshake flood
forged ClientHellos → **CPU budget** for failed checks